

Improving Diversity in LLM Short Story Generation

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Abstract

Large language models (LLMs) can generate accurate responses, but these are void of diversity. We attempt to address this for the task of creative short story generation. Drawing on established writing conventions and known LLM limitations, we target variation in genre, tone, style, and named entities. To promote diversity across these dimensions, we introduce DivLM, an LLM post-training framework consisting of two phases. First, we perform continued pre-training on a creative writing corpus and restore instruction-following capabilities using weight residuals. We then apply reinforcement learning with a custom, composite reward function that jointly maximizes diversity across the targeted narrative dimensions while maintaining response quality. Our empirical results on two LLM families show that DivLM increases diversity metrics by more than 9% on average compared to alternative approaches, while preserving instruction following, overall response quality, and similarity to human outputs.

Source code: github.com/ZahraSolatiDehkordi/DivLM

1 Introduction

Large language models (LLMs) have significantly improved in their ability to follow instructions and produce accurate and helpful responses. These advances have led to growing interest in using LLMs for creative writing and storytelling (Pei et al., 2024; Venkatraman et al., 2025; Wang et al., 2025a; Liu et al., 2024). However, prior studies show that while LLM assistance can increase the perceived creativity of individual human-authored texts, it reduces the collective diversity of content produced by groups of writers (Doshi and Hauser, 2024; Padmakumar and He, 2024). This concern is consistent with evidence that LLMs often exhibit limited diversity across repeated generations

from the same or similar prompts (Li et al., 2025b; Chakrabarty et al., 2024; Chu et al., 2025; Yun et al., 2025; Wang et al., 2025b; Xu et al., 2025). This low diversity is largely attributed to the post-training process, where LLMs are trained using methods such as instruction tuning and reinforcement learning from human feedback (RLHF) to increase output quality and alignment. This can narrow a model’s output distribution toward high-reward or typical responses (Padmakumar and He, 2024; Yun et al., 2025; Kirk et al., 2024). Such distributional narrowing can lead to mode collapse, where a model repeatedly generates outputs from a limited subset of plausible responses (Hamilton, 2024). This limitation is particularly important in creative writing workflows, where users often rely on LLMs to explore a space of possible ideas.

A common way to encourage more varied outputs is to increase sampling stochasticity through decoding methods such as temperature scaling. However, prior work has found that such methods do not necessarily improve diversity while coherence or quality drops (Peeperkorn et al., 2024; Chu et al., 2025). Other existing approaches include prompt-based (Wang et al., 2025b; Chu et al., 2025; Zhang et al., 2026a) or post-training (Lanchantin et al., 2025; Li et al., 2025c,b; Yao et al., 2025; Chung et al., 2025; Chen et al., 2026) methods. Our work rather improves the diversity of already post-trained models. Specifically, we focus on short story generation and adopt a broad view of diversity that considers higher-level narrative decisions. Figure 1 illustrates why this broader view is necessary. Generations may differ in surface details, but maintain the same underlying narrative trajectory and other high-level attributes. We define diverse short story generation as the task of producing a set of high-quality stories for a single prompt such that they differ along salient narrative dimensions. This moves past conventional single-output generation, where the objective is to produce

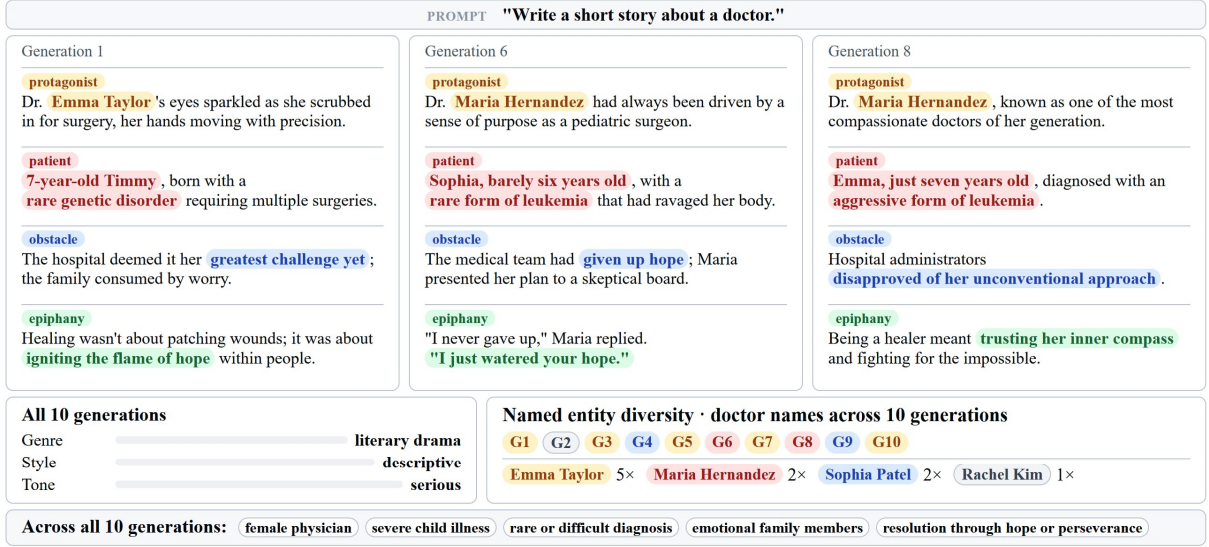


Figure 1: Diversity absence (mode collapse) for Llama-3.1-8B-Instruct across generations for the same prompt.

one fluent and instruction-following response. To reinforce this set property, we propose an optimization process that is based on specific group-level rewards computed over multiple generations for the same prompt. This diversification methodology is captured by a two-phase training framework; we refer to it as **Diversified Language Model** or **DIVLM**. First, we perform continued pre-training on creative writing data to strengthen the model’s domain-specific generative capabilities. Then, we apply reinforcement learning with carefully curated group-level rewards that encourage diversity while preserving basic output quality. Our approach promotes diversity over story-level attributes (genre, style, tone, named entities) and maintains typical quality standards. Empirically, we find that optimizing story-level diversity improves not only the targeted attributes, but also embedding-based semantic distance. Our main contributions are summarized as follows:

1. We propose a two-phase training framework (DIVLM) that combines continued pre-training on creative writing data with reinforcement learning based on group-level diversity rewards.
2. We introduce a composite reward formulation that promotes diversity over story-level attributes, including genre, style, tone, and named entities, while maintaining basic output quality.
3. Our analysis introduces novel metrics for quantifying diversity in LLM outputs. These go beyond cosine distance and are better tailored for the specificity of the short story generation task.
4. Our extensive experiments showcase that DI-

VLM yields significant gains on diversity performance while maintaining response quality.

2 Related Work

Prior work on LLM diversity can be broadly divided into methods applied during either model training or inference. The latter encourage diverse generations without updating model parameters, either by changing the decoding procedure or by modifying the prompt. Decoding-based approaches modify the token selection process during generation, aiming to avoid overly deterministic or repetitive outputs while preserving relevance and coherence (Franceschelli and Musolesi, 2025; Troshin et al., 2025). Prompt-based approaches instead induce diversity by changing the input context or task framing (Wang et al., 2025b; Chu et al., 2025; Zhang et al., 2026a). While these methods intervene at inference time to induce more varied outputs, they do not directly modify the model’s learned generation behavior. Training-time methods, on the other hand, incorporate diversity as an auxiliary objective during post-training procedures such as fine-tuning, reinforcement learning, and preference optimization. These approaches differ in how they define the diversity signal. One line of work uses lexical or token-level proxies, encouraging variation in surface form or in the model’s token distribution (Li et al., 2025c; Lanchantin et al., 2025; Yao et al., 2025). A second line of work measures diversity in a semantic representation space, encouraging generated responses to be farther apart according to embedding-based

criteria (Li et al., 2025b; Chung et al., 2025; Chen et al., 2026). These approaches move beyond lexical variation, but they typically compress diversity into a single semantic distance or equivalence signal. The most closely aligned work to ours uses reinforcement learning to encourage more diverse selections (Anschel et al., 2025). Here, diversity is defined over the distribution of selected options rather than over token choices or embedding distances between free-form generations.

Despite this progress, existing diversity objectives remain limited for creative writing. Lexical and token-level metrics primarily capture surface variation, while embedding-based distances often collapse distinct creative dimensions into a single semantic similarity score. As a result, these approaches do not fully capture aspects of diversity that are central to creative generation. Moreover, much of the training-time literature focuses on preserving the base model’s diversity during post-training, whereas our work treats diversity as the primary target for improvement.

3 Problem Formulation

Let $\mathcal{P} = \{p_1, \dots, p_N\}$ denote a set of N story prompts. For each prompt p_j , a policy model π_θ generates a group of $G \geq 1$ outputs, $\mathcal{O} = \{o_1, \dots, o_G\}$. Each output o_i is a short story conditioned on p_j and obtains a reward score $R(o_i) \in [-1, 2]$, the derivation of which is detailed in section 4.2. The objective is to optimize π_θ such that the generated group of outputs exhibits high diversity while maintaining acceptable quality. To this end, we treat diversity as the primary objective and enforce quality as a minimum requirement through the reward design. A condensed version of the optimization objective is given by

$$\arg \max_{\theta} \mathbb{E}_{p_j \sim \mathcal{P}, \mathcal{O} \sim \pi_{\theta'}(\mathcal{O}|p_j)} \left[\frac{1}{G} \sum_{i=1}^G \Lambda_i \right], \quad (1)$$

where θ' denotes the previous policy parameters, and $\Lambda_i \sim R(o_i)$.

4 Improving Diversity in Creative Writing with DIVLM

The proposed methodology (DIVLM) consists of two phases (Figure 2). First, we perform continued pre-training to strengthen the model’s foundational capabilities in creative writing. We then apply reinforcement learning to encourage greater output diversity while maintaining core requirements such

as prompt adherence and completeness.

4.1 Continued Pre-training For Creative Writing (CPT Phase)

The objective of this phase is to expose the model to diverse creative writing examples so that it can more effectively generate varied and imaginative outputs in subsequent tasks. This will also enhance the effectiveness of reinforcement learning (section 4.2) which depends on the capacity of the base model (Guo et al., 2025) and can attain an enhanced performance ceiling via domain-specific continued pre-training (Liu et al., 2025). Hence, we devise a domain-specific corpus (denoted by $\mathcal{C}$) that consists of creative writing texts curated to encourage stylistic flexibility and narrative variety (section 5.1).

In our experiments, we perform continued pre-training using a low-rank adapter (LoRA), keeping the underlying model weights fixed (Hu et al., 2022). In line with standard practice, this is applied to the pretrained base model (Wang et al., 2024; Lu et al., 2025). This process does not retain the instruction-following behavior of the instruction-tuned model (Zhang et al., 2026b), and hence these capabilities must be subsequently restored. To achieve this, we first compute an instruction residual $\Delta\theta$ by subtracting the parameters of the base model (θ_{base}) from the instruction-tuned ones (θ_{inst}) (Jindal et al., 2025):

$$\Delta\theta = \theta_{\text{inst}} - \theta_{\text{base}}. \quad (2)$$

Continued pre-training is then applied to the base model parameters, θ_{base} , yielding $\hat{\theta}_{\text{base}}$. Finally, $\Delta\theta$ is added back to the updated parameters to recover the instruction-following behavior. According to our empirical analysis, an important design choice is to determine the proportion of $\Delta\theta$ that should be applied in order to recover instruction-following capabilities while preserving the diversity gains achieved during continued pre-training. We introduce a tunable scaling parameter $s \in [0, 1]$ to control this. Therefore, the weights of the CPT model are given by

$$\hat{\theta}'_{\text{base}} = \hat{\theta}_{\text{base}} + s\Delta\theta. \quad (3)$$

4.2 Enhancing Diversity with Reinforcement Learning (RL phase)

We employ Group Relative Policy Optimization (GRPO; Shao et al., 2024) on $\hat{\theta}'_{\text{base}}$ (Eq. 3) to enhance output diversity. GRPO’s objective aligns with the fact that diversity in LLM outputs can only

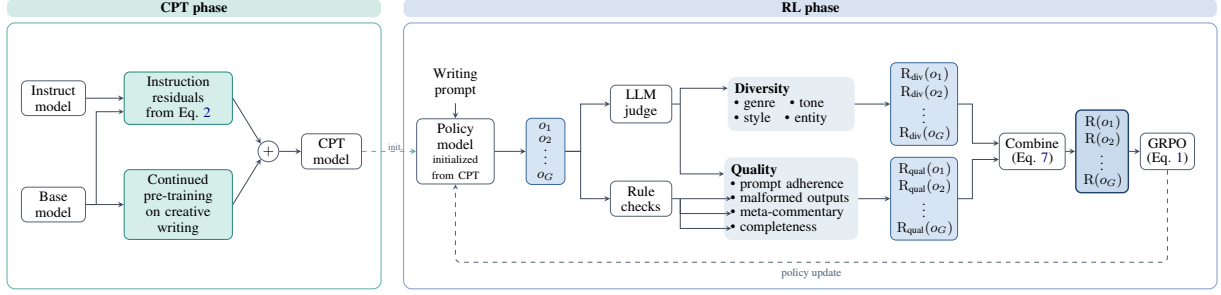


Figure 2: Overview of DIVLM, our proposed framework for improving diversity in short story generation, consisting of the CPT (section 4.1) and RL (section 4.2) phases.

be assessed by considering multiple responses per prompt. Furthermore, it reduces the critic-related memory and optimization overhead associated with Proximal Policy Optimization (PPO) style methods, as it foregoes a separately trained value model and directly exploits group-level reward signals (Shao et al., 2024). For each prompt p_j , GRPO samples a group of outputs $\mathcal{O} = \{o_1, \dots, o_G\}$ from the previous policy $\pi_{\theta'}$. The exact optimization task requires setting

$$\Lambda_i = \min \left(\rho_i A_i, [\rho_i]_{\phi}^{\psi} A_i \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{rf}}), \quad (4)$$

in Eq. 1, where π_{θ} and π_{rf} denote the current and reference (initialized from our CPT model) policy, $\rho_i = \pi_{\theta}(o_i | p_j) / \pi_{\theta'}(o_i | p_j)$, and $[\rho_i]_{\phi}^{\psi}$ denotes clipping from $\phi = 1 - \epsilon$ to $\psi = 1 + \epsilon$, wherein ϵ is a clipping hyperparameter. The advantage A_i is the standardized reward $R(o_i)$ calculated based on the reward values of all the outputs in the group. Finally, $\mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{rf}})$ penalizes divergence of the current from the reference policy, with the β hyperparameter controlling the strength of this regularization. Our reward formulation incorporates two components. The first focuses on **quality** requirements such as prompt adherence and completeness, while the other one promotes output **diversity**.

Quality Component. We combine both LLM-based evaluations and heuristic checks to provide a comprehensive quality assessment. We consider prompt adherence (R_{adhere}), absence of meta-commentary (R_{meta}), and narrative completeness (R_{comp}) as binary indicators, assigning scores that either 1 (true) or 0 (false). Further details are provided in Appendix D. The initial reward score, R_{qual} , for each generated output, o_i , is given by the average of the aforementioned three quality subscores. Outputs which are critically flawed, however, are assigned a fixed penalty that sets $R_{\text{qual}}(o_i) = -1$. The determination of this is based

on structural and linguistic quality criteria, such as length and coherence (Appendix D). This serves as a hard constraint to prevent the policy from reward hacking through trivially short or malformed text. Thus, the quality reward $R_{\text{qual}}(o_i) \in \{-1, 0, \frac{1}{3}, \frac{2}{3}, 1\}$.

Diversity Component. The second component of the proposed reward function encourages diversity in generated short stories. While reference-based metrics to measure lexical and semantic diversity can be used, they are vulnerable to reward hacking (Skalse et al., 2022; Pan et al., 2021). Moreover, prior work has shown that they often fail to capture nuanced, open-ended qualities such as style and narrative variation (Li et al., 2025a). We therefore only use them for evaluation and not as reward signals in GRPO training.

Our reward function adopts a more abstract representation of narrative diversity that focuses of four dimensions: genre, style, tone, and named entities (including main characters). We deploy an LLM-as-a-judge to determine those. For each one of genre, style, and tone diversity, we obtain a reward score as follows. Suppose we consider K subtypes or categories (e.g., different genres). In order to achieve a uniform distribution of these subtypes that maximizes diversity across the G prompt outputs/stories, each one must appear G/K times, assuming that $\gamma = G \bmod K = 0$. If $\gamma > 0$, the closest distribution to a uniform one will have $c_L = \lfloor G/K \rfloor$ instances of $K - \gamma$ subtypes and $c_H = \lceil G/K \rceil$ instances of γ subtypes. For an output o_i , we count the amount of times, $\xi \leq G$, its subtype appears in the G outputs. We then compute a proximity score, $S(o_i) \in [0, 1]$, which is a normalized deviation of ξ from a uniform subtype distribution. It is given by:

$$S(o_i) = 1 - \min(|\xi - c_L|, |\xi - c_H|) / d_{\text{max}}, \quad (5)$$

where $d_{\text{max}} = G - c_H$ is the largest possible devia-

tion. Let $\eta \leq K$ denote the number of subtypes in the G outputs that have a count equal to c_H , and ζ denote the number of subtypes in the G outputs that have a count equal to c_L . In the ideal case, $\eta = \gamma$ and $\zeta = K - \gamma$. When $\gamma > 0$, for outputs that are allocated a genre with total count c_H , their scores are corrected by applying a penalty $P(o_i) = \max(0, (\eta - \gamma)/K)$, where $P(o_i) \in [0, (K - \gamma)/K]$. Similarly, for outputs that are allocated a genre with total count c_L , their scores are corrected by applying a penalty $P(o_i) = \max(0, (\zeta - K + \gamma)/K)$, where $P(o_i) \in [0, \gamma/K]$. Thus, the final reward is calculated as $S(o_i) - P(o_i)$. Since the penalty $P(o_i)$ is only applied when $S(o_i) = 1$, the reward lies in $[0, 1]$. Rewards are calculated separately for genre, style, and tone, denoted by R_{genre} , R_{style} , and R_{tone} .

Named entity diversity operates over an open-ended rather than a fixed set of possible categories, and therefore requires a different reward formulation. We encourage named entities in each output o_i to differ as much as possible from those in other outputs within the same group (G outputs in total). Suppose that o_i has a set of named entities $\mathcal{Q}_i$. The named entity reward $R_{\text{ne}}(o_i) \in [0, 1]$ is defined by

$$R_{\text{ne}}(o_i) = \left(1 - \frac{1}{|\mathcal{Q}_i|} \sum_{\nu \in \mathcal{Q}_i} \frac{g(\nu) - 1}{G - 1} \right)^h, \quad (6)$$

where $g(\nu)$ counts the number of outputs ($\leq G$) that contain entity $\nu \in \mathcal{Q}_i$, and $h > 0$ is a tunable parameter that controls the sensitivity of the reward. To prevent the model from exploiting R_{ne} by producing a disproportionally large amount of named entities, we threshold both the denominator and the summation term of Eq. 6 to a value $u \in \mathbb{N}$ which is equal to the expected amount of entities produced by the original instruct model (θ_{inst}) on a held-out validation set. Thus, we replace $|\mathcal{Q}_i|$ with $\min(|\mathcal{Q}_i|, u)$ and $\sum_{\nu \in \mathcal{Q}_i} \frac{g(\nu) - 1}{G - 1}$ with $\min\left(\sum_{\nu \in \mathcal{Q}_i} \frac{g(\nu) - 1}{G - 1}, u\right)$.

This calibrates the clipping threshold to the baseline entity count, preventing outputs with unusually many named entities from weakening the normalization effect. For outputs that contain no named entities ($|\mathcal{Q}_i| = 0$), we assign a neutral score, equal to the mean R_{ne} across the outputs that contain at least one. Thus, the model is not encouraged to omit or introduce named entities unnecessarily. When no named entities exist in the G outputs, we assign $R_{\text{ne}}(o_i) = 0.5$ to all, so that the entity diversity reward introduces no relative preference among them.

Combining all the above steps, the total diversity reward for an output o_i , $R_{\text{div}}(o_i) \in [0, 1]$, is obtained by computing the average across $R_{\text{genre}}(o_i)$, $R_{\text{style}}(o_i)$, $R_{\text{tone}}(o_i)$, and $R_{\text{ne}}(o_i)$.

Total RL Reward. The final reward combines the quality and diversity components as follows:

$$R(o_i) = \begin{cases} R_{\text{qual}}(o_i) & R_{\text{qual}}(o_i) < \tau \\ R_{\text{qual}}(o_i) + R_{\text{div}}(o_i) & \text{otherwise,} \end{cases} \quad (7)$$

where $\tau \leq 1$ is a predefined quality threshold calibrated on a validation set to reflect the minimum quality acceptable for the task. As discussed previously, $R_{\text{qual}}(o_i) \in [-1, 1]$ and $R_{\text{div}}(o_i) \in [0, 1]$. Therefore, $R(o_i) \in [-1, \tau)$ when $R_{\text{qual}}(o_i) < \tau$ and $R(o_i) \in [\tau, 2]$ otherwise. Thus, overall $R(o_i) \in [-1, 2]$. The formulation in Eq. 7 ensures that diversity is only rewarded once a minimum level of structural quality is achieved. As noted by Yao et al. (2025), incorrect outputs may artificially increase their reward by exploiting the diversity term. By withholding the diversity reward when quality is below the threshold, we prevent favoring low-quality but superficially diverse outputs. If $R_{\text{qual}}(o_i) < \tau$, the model is trained based only on a quality-driven signal. Importantly, this does not impose a hard penalty (e.g., zero reward), but instead provides a graded signal that enables incremental improvement in output quality. When the threshold is reached, the diversity reward $R_{\text{div}}(o_i)$ is added to the total reward. This encourages the exploration of diverse outputs while still receiving quality signals.

This design naturally induces a form of curriculum learning (Bengio et al., 2009). Early in training, when outputs are more likely to fall below the threshold, the model optimizes a simpler objective focused on quality. As it learns to consistently exceed the threshold, the objective expands to include diversity, resulting in a richer optimization target. This staged objective decomposition is consistent with recent approaches in which prerequisite behaviors are learned before more complex objectives are optimized within the same training pipeline (Gao et al., 2024; Jiang et al., 2025).

5 Experimental Results

We use two open-source instruction-tuned models, Llama-3.1-8B (Grattafiori et al., 2024) and Gemma-2-9B (Gemma Team, 2024). We first describe the datasets and evaluation metrics, then analyze the effects of the two training phases (CPT and RL) as well as provide further insights via ablation.

5.1 Datasets

For the CPT phase (Section 4.1), we constructed an 18M-token corpus consisting of long-form creative writing from Project Gutenberg¹ and BookCorpus (Zhu et al., 2015). Original corpora from both resources include full-length novels or stories across multiple genres and historical periods. To transform this material into a high-quality training corpus, we deployed a multi-stage curation pipeline consisting of language filtering, toxicity screening, and deduplication (see Appendix C.1). For the RL phase (Section 4.2) as well as for the evaluation of all methods, we use the WritingPrompts dataset that contains human-written prompts and short stories (Fan et al., 2018). Original data consists of posts (used as prompts) and their associated comments, each one being a story. Each prompt or story has a score that reflects community reception, calculated as the sum of all upvotes and downvotes received from community members. We filtered the dataset based on this score. Additionally, we removed prompts with high semantic similarity, and applied balanced sampling across implied genres, styles, and tones (see also Appendix C.2).

5.2 Diversity Categories and Judge Validation

We have derived 9 genre categories from an empirical analysis of the book cataloging website Goodreads.² Style is represented using 3 modes: descriptive, action-focused, and dialogue-heavy, following prior narrative mode taxonomies and controllable generation settings (Selgin, 2009; Morrell, 2006; Marshall, 1998; Zhong et al., 2023); see also Appendix A. Tone (humorous, somber, neutral) captures the overall emotional register of the writing. Appendix B provides a more detailed overview of these categories.

To assess the reliability of the LLM judge, we compare its annotations with human ones on a subset of the data. The judge-human agreement rate is 89.86% which is quite similar to the human-human agreement of 91.63% (see Appendix G).

5.3 Evaluation Metrics and Experimental Settings

We evaluate generated outputs using metrics that quantify quality and diversity. All metrics are in the range of $[0, 1]$. Diversity metrics are computed at

the output group level (G outputs for each prompt). We report both the RL reward and the coverage of genre, tone, and style. Coverage is defined as the average proportion of subtypes that are present in each group of outputs. We also report the named entity reward score. To compare with a conventional baseline metric, we include the average pairwise cosine distance between the embeddings of the outputs in each group. To ensure consistency with the scales of the other metrics, we rescale cosine similarity from $[-1, 1]$ to $[0, 1]$ and derive cosine distance from the rescaled values, resulting in a cosine distance scale of $[0, 1]$. Quality metrics are computed at the individual output level. They include prompt adherence, meta-commentary rate, completeness, and coherence (see Appendix D).

We create validation and test sets of 100 prompts each from the curated WritingPrompts dataset. We select prompts that maximize semantic diversity and maintain balanced representation across implied genres, styles, and tones (see Appendix C). We use the validation set to select the hyperparameters and configuration settings for our experiments. CPT is performed on the corpus described in Section 5.1, while reinforcement learning is performed using 3,000 training prompts with a group size of $G = 10$. We tune the scaling parameter $s \in [0, 1]$ on the validation set by evaluating values in increments of 0.1. The named entity sensitivity parameter $h \geq 1$ is evaluated over $h \in \{1, 2, 3, 4, 5\}$. The quality threshold τ can take values in $\{-1, 0, 1/3, 2/3, 1\}$, of which we evaluate $\tau \in \{2/3, 1\}$. Because $\tau = 1/3$ permits two of the three quality metrics to fail, we exclude $\tau \leq 1/3$ as overly lenient. Based on the combined quality and diversity reward on the validation set, we select $s = 0.6$, $\tau = 2/3$, and $h = 5$ for the final experiments. We evaluate each model on the 100 test prompts, generating $G = 10$ outputs per prompt, for a total of 1,000 outputs per model. Additional configuration details are provided in Appendix H.

5.4 Model Performance

Table 1 enumerates the test performance metrics for the baseline instruct models (INS), compared to results obtained after CPT, and the final outcomes that encompass both phases of our proposed DivLM. We also provide comparisons with other methods from the literature, namely Diversity Quality Optimization (DQO; Chen et al., 2026) and Diversified Direct Preference Optimization (DDPO; Chung et al., 2025) using our datasets

¹Project Gutenberg, [gutenberg.org](https://www.gutenberg.org)

²Goodreads, [goodreads.com](https://www.goodreads.com)

Table 1: Comparison of the original Instruct (INS) LLM variants with CPT-tuned and DivLM (shortened to DLM) models across quality and diversity metrics ranging in $[0, 1]$. For Llama-3.1-8B, we include DQO and DDPO. To allow the calculation of meaningful averages, we report the no meta-commentary rate. Greater ($\uparrow$) values indicate better performance.

	Metric	Llama-3.1-8B					Gemma-2-9B		
		INS	CPT	DQO	DDPO	DLM	INS	CPT	DLM
Quality	Coherence	.997	.999	.999	.886	.997	1	.998	.995
	Prompt adherence	.997	.995	.999	.834	.981	.999	.998	.990
	No meta-comments	.968	.865	.952	.399	.998	.983	.956	.998
	Completeness	.726	.111	.983	.017	.985	.633	.128	.979
	AVG	.922	.742	.983	.534	.990	.904	.770	.990
Diversity	Genre coverage	.210	.270	.226	.341	.382	.224	.265	.312
	Tone coverage	.603	.717	.572	.853	.832	.657	.761	.741
	Style coverage	.542	.684	.542	.657	.801	.542	.623	.811
	AVG	.452	.557	.447	.617	.672	.474	.550	.621
	$\mu(R_{\text{genre}})$	.293	.413	.354	.567	.606	.322	.417	.506
	$\mu(R_{\text{tone}})$	.345	.532	.336	.698	.667	.429	.567	.542
	$\mu(R_{\text{style}})$	.291	.546	.294	.435	.644	.296	.454	.593
	$\mu(R_{\text{ne}})$	.581	.788	.535	.852	.902	.676	.794	.806
	AVG	.378	.570	.380	.638	.705	.431	.558	.612
	Cosine distance	.125	.174	.135	.236	.226	.124	.152	.179

and Llama-3.1-8B (see Appendix H).

CPT models outperform INS models across all diversity metrics. Prompt adherence and coherence remain comparable, while completeness decreases substantially, resulting in a lower average quality score. These results suggest that while CPT broadens the model’s output distribution, the subsequent reinforcement learning phase is necessary to recover output quality.

DivLM models yield quality metrics that remain close to those of the INS models, while completeness improves substantially. All diversity metrics improve. RL rewards respectively increase on average by 86.5% and 42% for Llama and Gemma compared to their INS baselines; respective gains over CPT are 23.7% and 9.7%. Smaller gains for Gemma may be partly due the fact that the hyperparameters were only optimized for Llama.³ Nevertheless, the consistent improvement observed for Gemma suggests that the approach transfers across model families. A similar pattern of gains is observed in the coverage metrics, where CPT improves INS by 23.23% and DivLM improves over CPT by 20.65%.

The cosine distance also increases, which indicates greater semantic variation despite not being directly optimized during training. The human-

³To accommodate experiments under tight computational cost (GPU hours) constraints.

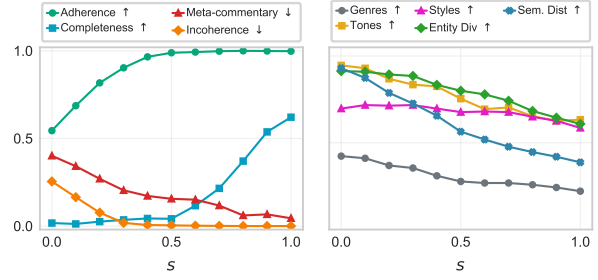


Figure 3: Quality (left) and diversity (right) evaluation results for increasing proportions of the instruction residual ($\Delta\theta$) using Llama-3.1-8B.

written stories associated with the test prompts have an average cosine distance of .297. Relative to this human reference, the Llama INS model achieves 42.1% of the observed semantic variation, while Llama DivLM reaches 76.2%; for Gemma, this increases from 41.8% to 60.4%. Hence, DivLM substantially narrows the gap between human and LLM-generated outputs.

From the baselines, the DQO model achieves high output quality, with an average score of .983. Compared with the INS model, it exhibits greater genre diversity and cosine distance but lower tone and named entity diversity. In contrast, the DDPO model improves across all diversity metrics relative to the INS model and achieves the highest cosine distance among all evaluated models. However, these gains come at a substantial cost to output quality, with DDPO obtaining the lowest average quality score of .534 and a completeness rate of only .017. Compared to DQO, DivLM improves quality by 0.71%, genre/tone/style coverage by an average of 50.34%, and diversity rewards by 85.53%. The respective gains compared to DDPO are 85.39%, 8.91%, and 10.50%.

5.5 Instruction Residual Scaling Analysis

Using Llama-3.1-8B, we assess the performance of CPT models that use an increasing amount of the instruction residual ($\Delta\theta$), controlled by $s \in [0, 1]$ in increments of .1. The results (Figures 3, S3, and S4) show a clear trade-off between output quality and diversity. As s increases, adherence and completeness improve substantially, while meta-commentary and incoherence decrease. Adding more of the instruction residual makes the model more reliable, but also shifts it back toward the less diverse behavior of the INS model.

Continuing this experiment into the RL phase, we observe that lower values of s result in lower

Table 2: Ablation study based on Llama-3.1-8B. We compare the following model variants: Instruct (INS), GRPO-only, DLM_{cos}, and DivLM (shortened to DLM) across quality and diversity metrics ranging in [0, 1]. Greater ($\uparrow$) values indicate better performance.

		Llama-3.1-8B			
		INS	GRPO	DLM _{cos}	DLM
Quality	Coherence	.997	.999	.999	.997
	Prompt adherence	.997	.996	.987	.981
	No meta-commentary	.968	.996	.999	.998
	Completeness	.726	.986	.850	.985
	AVG	.922	.994	.959	.990
Diversity	Genre coverage	.210	.250	.319	.382
	Tone coverage	.603	.603	.616	.832
	Style coverage	.542	.582	.539	.801
	AVG	.452	.478	.491	.672
	$\mu(R_{\text{genre}})$	.293	.383	.546	.606
	$\mu(R_{\text{tone}})$	.345	.345	.360	.667
	$\mu(R_{\text{style}})$	.291	.349	.237	.644
	$\mu(R_{\text{ne}})$	.581	.712	.800	.902
	AVG	.378	.447	.486	.705
	Cosine distance	.125	.167	.230	.226

initial quality scores, making it more difficult for the model to reach and maintain the quality threshold (Figure S5). For $s \in [0.0, 0.4]$, the model either never reaches the threshold or fails to remain above it and therefore does not sustain training for diversity. For $s \in [0.5, 0.9]$, the model eventually crosses and maintains the quality threshold, allowing it to train successfully for diversity. At $s=1$, the model begins with the highest quality score; however, it is significantly more constrained, leaving insufficient flexibility for diversity optimization. Therefore, our empirical analysis indicates that a reasonable validation space for s is in the $[0.5, 0.9]$ range.

5.6 Ablation Study

To assess the contribution of each component of our methodology, we evaluate two ablated variants. First, we apply GRPO training (RL phase) directly to the INS model without conducting CPT (GRPO-only). We also train an alternative DivLM variant in which our custom diversity reward is replaced with pairwise cosine distance (DLM_{cos}). The results are enumerated in Table 2. Complementary ablation analyses that assess different values for the group size, the decoding temperature, and the sensitivity parameter in the named entity diversity reward (h in Eq. S4) are provided in Appendices L, K, and J, respectively.

The GRPO-only model marginally achieves the

highest average quality score. This can be explained by the fact that training begins directly from the INS model, which already has strong quality performance, and the GRPO-only model primarily needs to improve completeness and reduce meta-commentary. However, GRPO-only produces the smallest diversity gain among the trained variants, increasing the average diversity reward by only 18.3% relative to the INS baseline. This indicates that GRPO can improve diversity without CPT, but its effect is limited when optimization begins from the more constrained INS model. Our interpretation is that CPT expands the set of responses available to the model, giving GRPO a broader space of trajectories for optimization.

The DLM_{cos} baseline improves average diversity rewards by 28.6% relative to the INS model and achieves the highest semantic distance among all variants, as expected from its training objective. However, its average diversity reward remains below that of its CPT initialization. In particular, its tone and style rewards decrease substantially, suggesting that optimizing semantic distance alone does not reliably improve the broader range of diversity dimensions captured by our reward. Notably, the semantic distance achieved by this model is only 1.7% higher than that of our final model. This indicates that, despite not directly optimizing cosine distance, our model achieves comparable semantic variation while substantially improving diversity across the broader set of target dimensions.

6 Conclusion

This work addresses the limited diversity of LLM outputs in short story generation, with a focus on four narrative dimensions, namely genre, tone, style, and named entities. We introduce DivLM, a two-phase post-training framework that combines CPT with diversity-focused RL. In the CPT phase, we train the model on a creative writing corpus and use weight residuals to transfer instruction-following capabilities. We observe that the proportion of the residuals added controls the trade-off between output diversity and quality. During RL, we optimize the model using a custom composite reward that encourages variation across the four narrative dimensions while maintaining overall response quality. Our results show that DivLM improves diversity significantly both over the instruct model variant as well as competitive alternative ap-

proaches. Further analysis highlights the complementary roles of the two training phases, wherein CPT broadens the model’s output distribution, providing RL with a richer trajectory space to explore and optimize.

Limitations

This study focuses exclusively on short story generation and does not experiment with other forms of creative writing, such as long-form stories and extended role-playing sessions. These tasks introduce additional requirements such as narrative continuity and long-term memory, which may interact with diversity in ways that are not captured by our current evaluation framework. Future work could investigate whether the proposed training approach remains effective in longer and more interactive generation settings, and extend the reward design to consider these additional dimensions.

Due to computational and memory constraints, our experiments are limited to models with at most 9B parameters. Evaluating DIVLM on larger models would provide a more complete understanding of its effectiveness and generalizability.

Finally, this work addresses output diversity rather than creativity in its broader sense. Diversity involves the degree of variation across generated outputs and represents only one component of creative behavior. Developing reliable methods to evaluate and optimize creativity comprehensively remains an important direction for future work.

Acknowledgments

We would like to acknowledge the support from UKRI’s AIRR Gateway Programme providing access to GPU computational support via the Isambard AI cluster (project titled “Foundation models for sequential predictions”). Both authors would like to acknowledge the feedback provided by Jayroop Ramesh during various stages of this work. VL would like to acknowledge all levels of support from the UKRI-funded Science of Fundamental AI Research (SOFAIR) Lab.

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Appendix

A About Writing Styles

What we refer to as *style* is often termed as *narrative mode* in the creative writing and stylistics literature. Prior work has proposed several taxonomies of fiction-writing modes, which overlap substantially. Selgin (2009) identifies five modes: action, dialogue, thoughts, summary, and description, whereas Morrell (2006) expands to: action, exposition, description, dialogue, summary, and transition. Similarly, Marshall (1998) lists action, summary, dialogue, feelings/thoughts, and background as core modes. These taxonomies consistently distinguish between physical action, spoken dialogue, and narratorial or descriptive exposition, while differing mainly in whether internal thought, exposition, summary, and background information are treated as separate categories. To this end, we have chosen to follow the paradigm of prior work in controllable fiction generation that employs dialogue, action, and description as explicit writing modes (Zhong et al., 2023).

B Construction of Genre, Style, and Tone Dimensions

We construct the diversity dimensions (genre, style, and tone) to be as mutually distinguishable and semantically orthogonal as possible, while remaining interpretable to both human readers and LLM-based judges. Each dimension consists of a small set of discrete categories designed to support robust LLM judgement and balanced reward allocation.

Genre. The genre categories were derived empirically from an analysis of 10,634 books and their associated genres as reported on the book cataloging website Goodreads. We first removed all non-fiction genres (e.g., philosophy, religion, biography, politics). We then computed pairwise genre overlap and merged categories with high overlap (Jaccard index ≥ 0.2). Finally, we removed genres that primarily denote demographics or time periods (e.g., young adult, 20th century). The resulting set reflects broad narrative genres with minimal semantic redundancy: *fantasy*, *science fiction*, *horror*, *romance*, *mystery-thriller*, *literary drama*, *historical*, *action-adventure*, and *other*. The *other* category captures works that do not fit into the preceding genres or exhibit mixed or less common characteristics (e.g., dystopian fiction or magical realism).

Style (Narrative Mode). To reduce overlap and

improve robustness of automatic classification by an LLM judge, we collapse the finer-grained distinctions (see Appendix A) into three broad modes: *descriptive*, which includes narration, exposition, description, and background; *action-focused*, emphasizing physical events and movement; and *dialogue-heavy*, in which story progression occurs primarily through character speech. This formulation is consistent with prior work in controllable fiction generation, which uses dialogue, action, and description as controllable writing modes (Zhong et al., 2023). Although individual stories may exhibit elements of multiple modes, each story is assigned a single label that identifies its dominant narrative style.

Tone. We define tone using three categories: *humorous*, *somber*, and *neutral*. Rather than treating humor as a separate genre, we assume that tone is orthogonal to genre, allowing any genre to be expressed in a humorous, somber, or neutral manner.

These dimensions define a compact and interpretable space over which diversity can be enforced, addressing aspects of creative writing that are not always captured by surface-level similarity metrics.

C Corpora Preprocessing

C.1 Creative Writing Dataset

To form the dataset for continued pre-training, we first curate the selection of 150 books to ensure a uniform distribution across genres. Books are obtained from both Project Gutenberg and the BookCorpus (Zhu et al., 2015) dataset. We filter the books to retain only the narrative text of the story. We remove front and back matter (e.g., tables of contents), boilerplate (e.g., *** END OF THE PROJECT GUTENBERG EBOOK ***), illustration markers and captions (e.g., [Illustration]), and formatting artifacts, such as decorative separators (e.g., *****) and layout markers (e.g., /* RIGHT . . . */). We identify and remove these elements using rule-based pattern matching. The texts are segmented at sentence boundaries into passages of 500-800 tokens, with token counts measured using the Llama-3.1-8B tokenizer. Consecutive sentences are added until adding the next sentence would exceed the 800-token limit. We perform language filtering using the fastText language identification model (Joulin et al., 2016, 2017) and retain only English texts. We apply automated toxicity screening to the BookCorpus dataset using the

Detoxify library⁴ to identify passages containing toxic or otherwise unsuitable content. We set the threshold values for explicit, obscene, and toxic content to 0.2, 0.6, and 0.6, respectively. Any passage exceeding these conservative thresholds in the corresponding toxicity dimensions is removed. This process filters out undesirable material without constraining the creative or stylistic diversity of the corpus. This filtering not applied to books from Project Gutenberg. To remove redundant passages while preserving the natural variability of creative writing, we follow the approach of Lee et al. (2022) for deduplication. We use exact matching, substring overlap (via 50-character substrings), edit similarity (> 0.7) and Jaccard similarity (> 0.7) as primary signals, and add cosine similarity (> 0.9) as an additional safeguard for semantic duplicates. Although these thresholds are intentionally strict, the inherent diversity of creative prose means that only 0.6% of passages are removed in this step.

C.2 Writing Prompts Dataset

The writing prompts are obtained from the WritingPrompts (Fan et al., 2018) dataset. We first filter prompts based on the human-assigned popularity scores in the dataset. We assume that strong community engagement, in the form of upvotes and substantive story responses, serves as a proxy for prompt quality and narrative potential. Therefore, we retain prompts with a post score greater than 5, more than two comments, and at least one comment with a score greater than 2. To identify and remove similar prompts, we compute normalized sentence embeddings using all-mpnet-base-v2 (Song et al., 2020; Reimers and Gurevych, 2019) and compute pairwise cosine similarities between all prompt embeddings. For prompt pairs with cosine similarity greater than 0.55, we remove one prompt from the pair. Examples of semantically similar prompts identified during this stage are shown in Figure S1.

To prevent the dataset from being biased toward specific genres, tones, or styles, we design a balanced sampling pipeline. We prompt the judge LLM to classify prompts into one of three categories: constrained, implied, or open-ended. For prompts labeled as constrained or implied, the judge additionally assigns the corresponding genre, style, or tone labels. Finally, the judge is asked to assess prompt validity in order to remove prompts

Pair 1

(a) “9 out of 10 dentists recommend this toothpaste. You are the 1 out of 10 who doesn’t – and for a very good reason.”

(b) “You disagree with most of your co-workers and peers. You hate toothpaste. You are the 10th Dentist.”

Pair 2

(a) “You are an attorney. A client tells you he’s found a genie and wants your help in crafting a wish with no ironic loopholes the genie can exploit. Your retainer fee is the third wish.”

(b) “The genie has taken a liking to you. ‘Take this notebook. Make your wish as specific, loophole free as possible. I will return in one year.’ You immediately hire a lawyer.”

Figure S1: Examples of semantically similar prompts identified during deduplication.

that cannot be meaningfully answered by a language model. Examples of invalid prompts include instructions that depend on external or unavailable context, such as “Instead of this title, take any title from your Reddit frontpage as prompt” or “Write a story about this pic that made the front page of Reddit.” The final dataset is then constructed by uniformly sampling valid prompts across the identified genres, styles, and tones to ensure balanced representation.

D Quality Metrics

The quality metrics used during training are prompt adherence, meta-commentary, and completeness. We use Gemma-3-12b-it as a judge to evaluate these metrics (Gemma Team, 2025). The judge is prompted with a structured system message defining evaluation criteria and a required JSON output format, ensuring consistent and parsable responses. Prompt adherence assesses whether the generated story satisfies the essential requirements specified in the prompt, including characters, setting, constraints, and stylistic elements (if explicitly stated). Meta-commentary captures the presence of undesirable meta-textual content, defined as any content that falls outside the narrative itself. This includes statements about the story (e.g., “Here is your story”, “In this story I will”) and comments directed at the user. Completeness evaluates whether the story forms a coherent and self-contained narrative. A complete story should have a clear beginning, development, and resolution. Outputs are penalized if they are truncated or end abruptly (e.g., mid-sentence or mid-scene). In addition to

⁴Detoxify repository, github.com/unitaryai/detoxify

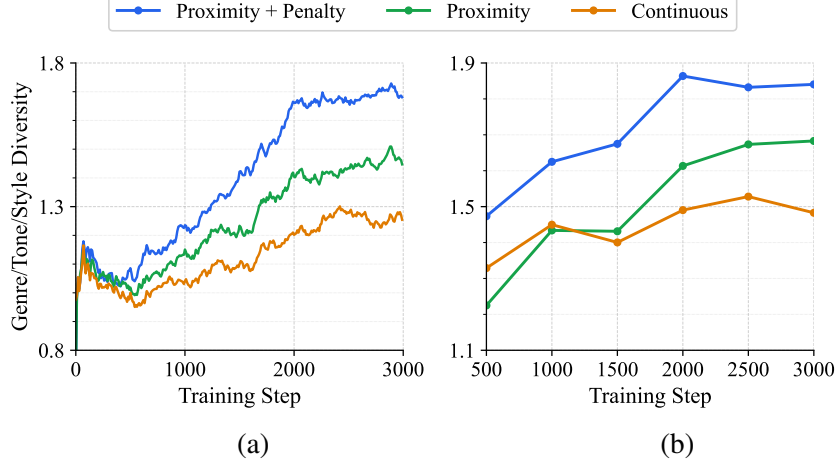


Figure S2: Genre, tone, and style diversity curves for different reward formulations on Llama. The score is the sum of the genre, tone, and style rewards, resulting in a range of $[0, 3]$. (a) Training scores. (b) Validation scores evaluated every 500 steps.

the LLM judge, we apply rule-based checks for meta-commentary and completeness. The meta-commentary check searches directly for common meta-textual phrases identified empirically, such as “Is there anything else I can help you with?”. The completeness check uses a simple punctuation heuristic, flagging outputs that do not end with a punctuation mark as incomplete. For each of these metrics, an output is marked as failing if either the LLM judge or the corresponding rule-based check detects a violation. All three metrics are binary, i.e., a score of 1 indicates the desired behavior, while a score of 0 indicates failure.

An output is classified as malformed if it violates basic structural or linguistic quality criteria. Specifically, we flag outputs that are insufficient in length (fewer than 50 words), exhibit low lexical diversity (a unique-token ratio below 30%), or contain formatting artifacts. Formatting artifacts include special token leakage (e.g., `<|im_end|>`, `<|endoftext|>`) as well as HTML or CSS fragments indicative of formatting corruption (e.g., `font-size`, `font-weight`). Furthermore, we prompt the judge to assign binary scores for coherence and grammatical correctness. Outputs receiving a score of 0 on either metric are counted as malformed.

E Alternative Formulations for the Diversity Reward

The diversity component of our methodology derives a diversity reward for genre, tone, and style (the reward for named entities is defined differ-

ently). We justify the choice of this reward, labeled as **Proximity + Penalty**, by comparing with two alternative, perhaps more straightforward formulations.

Proximity. This variant relies solely on the proximity score from Eq. 5:

$$R(o_i) = S(o_i). \quad (S1)$$

Continuous. This variant defines the ideal count as $c = G/K$, regardless of whether $G \bmod K = 0$. Thus, c may have a non-integer value, and the reward is computed as

$$R(o_i) = 1 - |\xi - c|/d_{\max}. \quad (S2)$$

We train the Llama model using each reward variant and report the training and validation curves (Figure S2) along with the final evaluation results (Table S1). The ‘Proximity + Penalty’ variant obtains the best genre, tone and style rewards.

F Named Entity Reward

We compare four formulations of the named entity reward. In all cases, $\mathcal{Q}_i$ denotes the set of normalized named entities extracted from output o_i , after removing any named entities that appear in the prompt. Named entities are extracted using our

Table S1: Reward performance of different diversity reward formulations on the test set. The score ($\in [0, 3]$) is the sum of the genre, tone, and style rewards.

Formulation	Score
Continuous	1.634
Proximity	1.712
Proximity + Penalty	1.917

Variant	Avg. unique entities per group	Avg. repeated entities per group
INS	28	4.7
Original	62	2.8
Exponential	21	2.1
Clipped	26	1.5
Penalty	49	2.4

Table S2: Results on the test set for different named entity diversity formulations with $G = 10$ on the Llama-3.1-8B model.

LLM judge model.

Original. This variant uses the reward from Eq. 6 without clipping.

Clipped. This variant applies the clipping described in Section 4.2 to Eq. 6. It is the reward used in our proposed model.

Penalty. This variant applies an entity count penalty to the original reward:

$$P_{ne}(o_i) = \exp\left(-\left|\frac{|Q_i| - \mu}{\sigma}\right|\right), \quad (S3)$$

where μ and σ are the mean and standard deviation of $|Q_i|$ estimated from the outputs of the INS model on the validation set. The penalty is applied by computing $R_{ne}(o_i) \cdot P_{ne}(o_i)$. This encourages the number of named entities to remain close to that of the baseline INS model, reducing the incentive to exploit the reward through unusually large or small entity counts.

Exponential. This variant removes the explicit normalization by $|Q_i|$ and instead applies an exponential transformation directly to the summation term:

$$R_{ne}(o_i) = \exp\left(-\sum_{\nu \in Q_i} \frac{g(\nu) - 1}{G - 1}\right). \quad (S4)$$

In all four variants, no-entity outputs are assigned neutral scores as described in Section 4.2. In the Penalty variant, $P_{ne}(o_i)$ is computed and applied to no-entity outputs after the neutral score is assigned. Table S2 reports the average number of unique and repeated entities per group on the test set for each variant. The original variant incentivizes the model to increase the number of unique entities, producing more than twice as many as the baseline. The penalty variant mitigates this behavior but does not fully prevent it. Overall, the clipped variant is the only formulation that keeps the number of unique entities close to the baseline while significantly reducing the number of repeated entities.

Metric	Human-human	Judge-human
Quality metrics		
Prompt adherence	92.30%	90.40%
Meta-commentary	97.60%	95.70%
Completeness	100.00%	94.10%
Coherence	97.00%	92.70%
Average	96.73%	93.23%
Diversity metrics		
Genre	80.60%	82.70%
Tone	84.60%	81.70%
Style	89.30%	91.70%
Average	84.83%	85.37%
Total average	91.63%	89.86%

Table S3: Human-human and judge-human agreement rates for quality and diversity metrics.

G LLM-as-a-Judge Evaluation Based on Human Labels

The use of LLM-as-a-judge is an active area of methodological debate. Prior work has shown that LLM judges can approximate human preferences at scale, but can also be vulnerable to issues such as position and verbosity bias (Zheng et al., 2023; Gu et al., 2025; Li et al., 2025a). To validate our evaluator, we manually annotate 100 samples, corresponding to 10% of the 1000-sample test set. We annotate all samples for prompt adherence, completeness, meta-commentary, genre, style, and tone. We report both inter-annotator agreement and human-LLM agreement (Table S3). Overall, human-human (91.6%) and judge-human (89.9%) agreement rates overlap significantly. Notably, inter-human agreement is substantially higher for the quality-related metrics than for the diversity-related metrics. This is expected, as judgments of genre, tone, and style are more subjective and may allow for several reasonable interpretations. For the quality-related metrics, judge-human agreement is slightly lower than inter-human agreement, indicating that the judge does not fully reproduce the relatively strong consensus among human annotators. In contrast, for the diversity-related metrics, judge-human agreement is slightly higher than inter-human agreement. Because judge-human agreement is measured against the aggregate human judgment, rather than against each individual annotator, aggregation can reduce annotator-specific variation. The judge may therefore agree more closely with the human consensus than individual humans agree with one another, particularly on the more subjective diversity-related metrics.

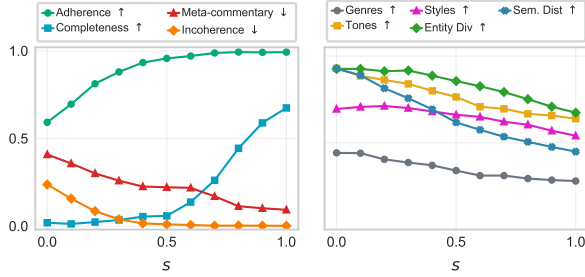


Figure S3: Validation results for different instruction residual scales s on Llama-3.1-8B. Diversity metrics consist of genre, tone, and style coverage values, as well as the named entity diversity reward R_{ne} . All metrics are reported on a $[0, 1]$ scale.

H Training Configuration

DIVLM. In the continued pre-training phase, we conducted training using LoRA ($r=32$, $\alpha=32$) applied to attention and feed-forward modules. Training was conducted for one epoch with a batch size of 16. The learning rate was set to 1×10^{-4} with cosine scheduling and 8% warmup. After adding the instruction residual, we applied GRPO to the resulting model using LoRA ($r=16$, $\alpha=16$) on the attention and feed-forward modules. The learning rate was set to 4×10^{-5} with cosine scheduling and 10% warmup. We conducted training for two epochs on 3,000 prompts with a batch size of 20, and a group size $G=10$. Across all experiments, the maximum generation length was set to 700 tokens and the temperature was set to 1. The continued pre-training phase was conducted on a single NVIDIA GH200 Grace Hopper Superchip, while the GRPO training was performed on four GH200 units. AdamW (Loshchilov and Hutter, 2019) optimizer was used in both training phases. The embedding model used to compute cosine distance was all-mpnet-base-v2 (Song et al., 2020; Reimers and Gurevych, 2019).

DQO. For DQO (Chen et al., 2026), we trained the Llama-3.1-8B INS model on our dataset of 3,000 samples using the authors’ released DQO-PPO implementation.⁵ We followed the authors’ reported hyperparameters where applicable: three training epochs, an actor learning rate of 1×10^{-6} , a critic learning rate of 1×10^{-5} , a rollout temperature of 1, Skywork-Reward-V2-Llama-3.2-1B as the reward model, and all-MiniLM-L6-v2 as the embedding model. To accommodate our larger model size, we reduced the batch size to 32. We set the

⁵DQO repository, github.com/fairytale9/diversity-quality-optimization

Table S4: Performance comparison between the GRPO and PPO variants of DQO.

Metric	GRPO	PPO
Average quality score	.894	.983
Average Genre/Tone/Style coverage	.454	.447
Average diversity reward	.360	.380
Cosine distance	.118	.135

maximum response length to 700 tokens for consistency with our other experiments. In addition to the PPO variant, we evaluated the GRPO variant of DQO. We selected the PPO variant for direct comparison with DIVLM as it achieved better overall results (Table S4).

DDPO. For DDPO (Chung et al., 2025), we trained Llama-3.1-8B using the authors’ released DiversityTuning implementation.⁶ Because the method requires human-written responses paired with the training prompts, we first retrieved the corresponding responses for our prompts. Following the authors’ pipeline, we fine-tuned the model on the resulting dataset. We then initialized DDPO from this fine-tuned checkpoint and trained it on the authors’ semantic and stylistic preference pairs constructed from the same dataset. Following the codebase, the DDPO stage used three training epochs, a learning rate of 5×10^{-6} , and $\beta=0.1$. We set the batch size to 4 with a maximum response length of 700 tokens.

I Instruction Residual Scaling Analysis

In addition to the results on the test set for different values of s (Figure 3), we report the corresponding results on the validation set in Figure S3 and the mean genre, tone, and style rewards on both the test and validation sets in Figure S4. The trends are consistent across all results, with quality improving and diversity decreasing as s increases. Figure S5 additionally presents the RL training curves for each value of s .

J Sensitivity Parameter (h) in Named Entity Diversity Reward

We experiment with different values of the sensitivity parameter h in the named entity diversity reward (Eq. 6). To facilitate comparison across reward scales, we report the percentage change in reward relative to its initial value and the average number

⁶Diversity Tuning repository, github.com/mj-storytelling/DiversityTuning

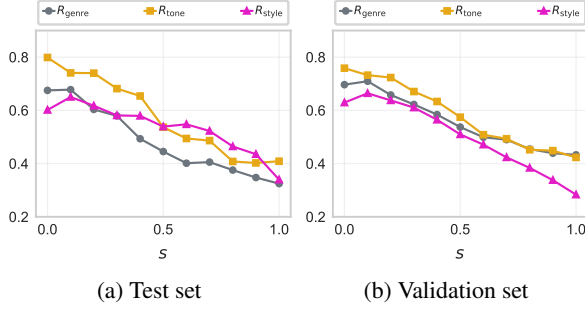


Figure S4: Genre, tone, and style rewards obtained for different instruction residual scales s on Llama-3.1-8B. Rewards are reported on a $[0, 1]$ scale.

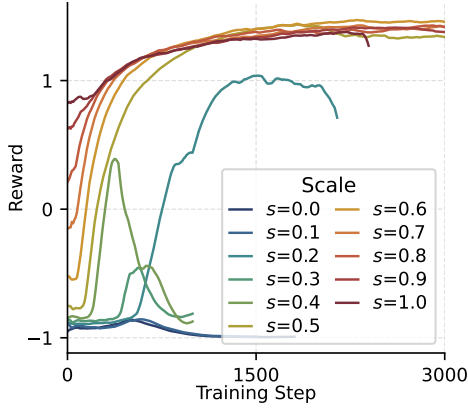


Figure S5: Training curves for Llama-3.1-8B under different values of s , showing the total reward $R \in [-1, 2]$ across training steps.

Table S5: Entity repetition ratio (repeated entities per group over total entities per group) and mean R_{ne} for INS and DivLM across values of h . Final evaluation uses $h = 5$ for consistency.

Metric	INS	$h=1$	$h=2$	$h=3$	$h=4$	$h=5$
Ratio $\downarrow$	.107	.117	.066	.068	.078	.053
$\mu(R_{ne}) \uparrow$	.581	.686	.838	.873	.813	.902

of repeated entities per group over training (Figure S6). At $h = 1$, the reward provides insufficient pressure to reduce entity repetition. Although it increases slightly early in training, this improvement is not sustained, and the number of repeated entities ultimately increases. As h increases, repeated entities are penalized more harshly, leaving greater scope for optimization. The best performance is achieved by setting $h = 5$, which obtains the largest improvement over the initial reward (55.25%) and the largest reduction in the number of repeated entities over total entities per group (50.47%) (Table S5).

Table S6: Average cosine distance on the test set for INS and DivLM under different values of T .

Model	$T=0.8$	$T=0.9$	$T=1.0$	$T=1.1$	$T=1.2$
INS	.116	.118	.125	.131	.138
DivLM	.215	.225	.226	.231	.236

K Decoding Temperature Analysis

The decoding temperature T is a hyperparameter that controls output randomness and is commonly used to increase diversity and creativity in LLM outputs (Peeperkorn et al., 2024; Troshin et al., 2025). It modifies the token probability distribution by scaling the logits before softmax normalization (Keskar et al., 2019; Renze, 2024). Values of $T > 1$ flatten the distribution, increasing the probability of sampling lower-ranked tokens. To evaluate the performance of our model under different decoding temperatures, we compute the diversity metrics on the test set for $T \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$ and compare the results with the INS model. Higher values of T lead to a similar increase in cosine distance for both the INS and DivLM models, while the gap between them remains consistent across the evaluated range (Table S6). Similar results are observed for the mean diversity reward (Figure S7). The DivLM model maintains substantially higher diversity at every temperature. Notably, the INS model at $T = 1.2$ does not approach the diversity reward and semantic distance achieved by the DivLM model at $T = 0.8$, which indicates that increasing decoding temperature alone is insufficient to obtain comparable levels of diversity. The complete diversity results for DivLM at different values of T are depicted in Figure S8). We observe a slight increase in named entity, style, and tone diversity at higher temperatures,

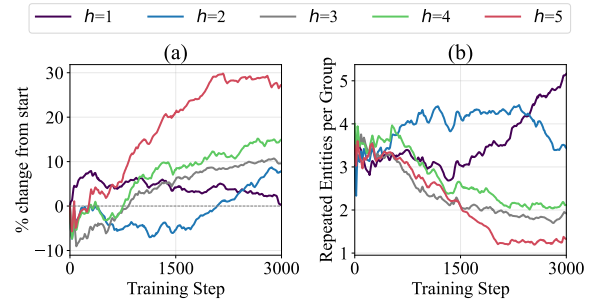


Figure S6: Training curves for Llama-3.1-8B under different values of h . (a) Percentage change in the entity diversity reward over training. (b) Average number of repeated entities per group over training.

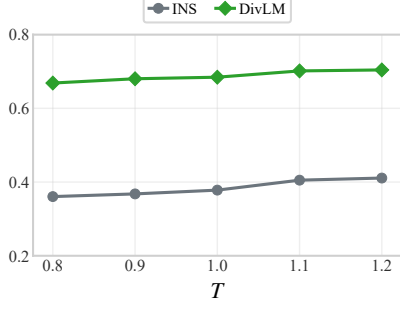


Figure S7: Mean diversity reward (R_{div}) on the test set for DivLM and INS across decoding temperatures T on Llama-3.1-8B.

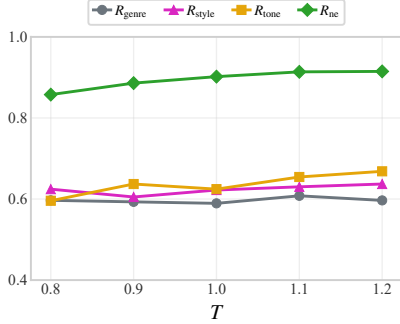


Figure S8: DivLM diversity evaluation results across decoding temperatures T on Llama-3.1-8B.

while genre diversity remains largely unchanged.

L Group Size Analysis

To evaluate the performance of our model with different group sizes, we compute the diversity metrics on the test set for $G \in \{5, 10, 20, 50, 100\}$ and compare with the INS model. The results show that cosine distance remains approximately constant across different group sizes (Table S7). The mean diversity reward obtained by DivLM maintains a consistent gap with INS (Figure S9). In

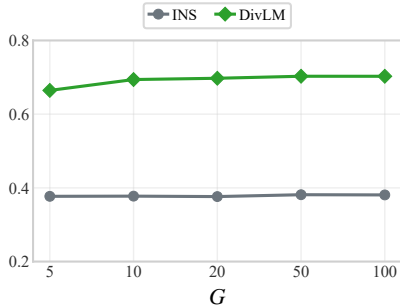


Figure S9: Mean diversity reward (R_{div}) on the test set for DivLM and INS across group sizes G on Llama-3.1-8B.

Table S7: Average cosine distance on the test set for INS and DivLM under different values of G .

Model	$G=5$	$G=10$	$G=20$	$G=50$	$G=100$
INS	.123	.125	.126	.126	.126
DivLM	.227	.226	.226	.226	.226

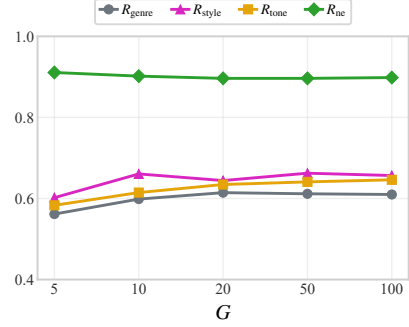


Figure S10: DivLM diversity evaluation results across group sizes G on Llama-3.1-8B.

addition, we report the complete diversity evaluation results of DivLM at different values of G (Figure S10). From $G=5$ to $G=10$, we observe slight increases in genre, tone, and style diversity, along with a slight decrease in named entity diversity. After $G=10$, all metrics stabilize.

M Length Penalty

We observe that RL training may result in shorter model outputs. Although our task does not have an ideal output length, keeping the average output length consistent is necessary for fair comparison, as output length can directly affect evaluation metrics. To mitigate this effect, we apply a length penalty to the GRPO reward with the goal of maintaining output lengths comparable to those of the INS model. An alternative approach would be to compute length-normalized metrics; however, metrics evaluated by LLM-based judges cannot be reliably length-normalized. Thus, we opt to explicitly regulate output length during training rather than normalizing evaluation metrics. We do acknowledge that variation in output length can itself be a form of diversity and may be sometimes desirable.

Setting	INS	DivLM
Decay = 50	481.2	366.3
Decay = 150	481.2	410.0
Linear	481.2	483.6

Table S8: Mean output length in words for the INS and DivLM models using different length penalties.

We first compute the average output length of the INS model, denoted by L_{avg} , and adopt this as the target average during RL. We compare exponential penalties with decay values of 50 and 150 against a linear penalty. In all cases, the length penalty lies in the range $[0, 1]$. For the linear penalty, we calculate the penalty as $\max(0, 1 - \frac{L_{o_i}}{L_{\text{avg}}})$, where L_{o_i} is the length of output o_i . Table S8 reports the mean output length under different penalty settings, showing that the linear approach preserves the length relative to the INS model. Accordingly, we adopt this formulation for our length penalty.